Christopher H. Vu

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Education

University of California, San Diego La Jolla, CA
B.S. Data Science, B.S. Mathematics-Computer Science *Exp. June 2028*

- **Relevant Coursework**:
  - **Core CS & Data**: DSC10, DSC20, DSC30, DSC40A, CS20, CS11 .
  - **Mathematics**: MATH180A, MATH181A, MATH189, MATH20A-D, MATH18

Projects

Predictive Modeling of High-Dimensional Competitive Match Data | *Python, ML, SQL* June 2025 – October 2025

- Designed and trained a gradient-boosted classifier (XGBoost) on 50,000+ high-Elo League of Legends matches, achieving **76% test accuracy** (outperforming the 50% random baseline).
- Engineered domain-specific features including **Elo-normalized synergy scores** and pairwise counter-deltas to quantify "Draft Advantage," increasing model AUROC by 8% over raw champion-select features.
- Constructed a robust ETL pipeline using SQLAlchemy and PostgreSQL to handle rate-limited API acquisition, data cleaning, and feature storage for high-dimensional match data.

Deep Convolutional GAN (DCGAN) Implementation | *PyTorch, Python, Computer Vision* December 2025 - Present

- Implemented a Deep Convolutional Generative Adversarial Network (DCGAN) from scratch in PyTorch, reproducing the architecture from *Radford et al. (2015)* to generate high-fidelity **28x28** synthetic MNIST digits.
- Stabilized adversarial training dynamics to reach a convergent Discriminator loss of **0.52** (near theoretical equilibrium) after **50 epochs**, effectively mitigating mode collapse and vanishing gradients.
- Engineered a Generator with transposed convolutions and a Discriminator with strided convolutions, replacing standard pooling layers to preserve spatial hierarchies and eliminate artifact noise.

Reinforcement Learning Blackjack Strategy Optimizer | *Python, Gymnasium, NumPy, Matplotlib* December 2025 - Present

- Architected a reproducible **Monte Carlo evaluation pipeline** to simulate 3,000+ training episodes and 5,000 evaluation hands with multi-seed variance tracking.
- Developed a **bet-aware Q-learning agent** featuring model serialization (`.pkl`) and a modular CLI for automated training and tournament execution (`train.py`, `tournament.py`).
- Quantified strategy risk/return profiles by computing distributional metrics (mean ± std); identified a performance gap where the learned policy ($-2,226.7 \pm 357.7$) trailed the flat-bet baseline (-387.7 ± 28.6).
- Implemented production-grade engineering standards including **headless plotting** (`matplotlib Agg`) and **CI-friendly integration tests** (11+ tests passing) to ensure system scalability.

CNN Architecture Benchmarking | *PyTorch, Computer Vision* September 2025 - October 2025

- Implemented and compared three distinct Convolutional Neural Network architectures (Custom Simple CNN, ResNet-18, EfficientNet-B0) to analyze performance/compute trade-offs.
- Applied **transfer learning** on pre-trained ResNet-18 weights, fine-tuning final fully connected layers to achieve >**95% accuracy** on binary classification tasks.
- Designed a reproducible training loop with learning rate scheduling, data augmentation (random crops/rotations), and model checkpointing to optimize convergence.

Experience

Strategic Performance Esports Coach June 2022 – Present
Santiago Canyon College / Freelance *Orange, CA*

- Coached League of Legends players by analyzing student's gameplay and match history to identify macro-strategic inefficiencies for competitive players (Masters+).
- Synthesized complex game states into actionable win-conditions, improving individual player rank consistency through data-informed decision making.

Vice President, Engineering Club September 2024 – June 2025
Santiago Canyon College *Orange, CA*

- Directed technical workshops and managed project timelines for Arduino-based team competitions.
- Oversaw resource allocation and club documentation, ensuring successful project delivery for student teams.

Technical Skills

Languages: Python, SQL, C++, C, Java, JavaScript, LaTeX, R, x86 Assembly
Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Big O Analysis
Machine Learning: PyTorch, Reinforcement Learning (Gymnasium), Computer Vision, Generative Models (GANs), XGBoost, Scikit-learn, NumPy, Pandas
Tools & Visualization: Git/GitHub, Linux/Unix, PostgreSQL, Matplotlib, Seaborn, SQLAlchemy